

Intelligent systems for industry, supply chain and environment

LESSON 4

What is intelligent? Machine learning basics,
Digital ages, Simulation of the exam



Outline

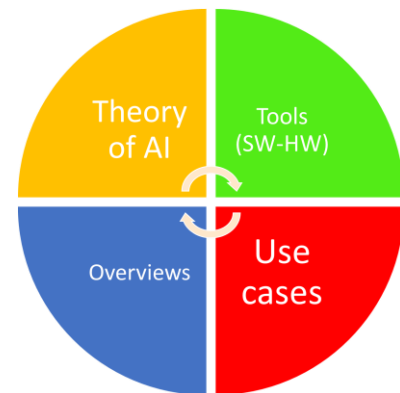
- Theory
 - What is intelligent?
 - Representation
 - Evaluation
 - Optimization
- Current and future scenarios where the intelligent systems play a relevant role
- Digital Ages
 - Collaborative age
 - Autonomous age
- Simulation #1 of the exam with comments
- **Main points**



THEORY

What is intelligent?

What are the features to call intelligent a system?



Artificial Intelligence (AI) vs Machine learning (ML)

- AI:
is the broader concept of machines being able to carry out tasks in a way that we would consider “**smart**”.
- ML:
a current application of AI based around the idea that we should really just be able to give machines access to data and let **them learn for themselves.**



Artificial Intelligent Systems

- One can debate endlessly about whether a certain system is intelligent or not.
- One good operational description is as follows
*“SW/HW systems designed to perform **complex tasks** employing strategies that mimic some **aspect of human thought**”*
- **The key is evolution**: it is intelligent if **it can learn** (even if only a limited sense) and/or get better in time

Is ML suitable for all applications?

There are some IFs...

IF

- the nature of computation required in the task is not well understood

OR

- there are too many exceptions to the rules

OR

- known algorithms are too complex or inefficient

THEN

- AI can be considered as a possible solution

ELSE

- Standard coding should be considered as the first option

We have to apply 3 concepts

Is ML suitable for all applications?

There are some IFs...

IF

- the nature of computation required in the task is not well understood

OR

- there are too many exceptions to the rules

OR

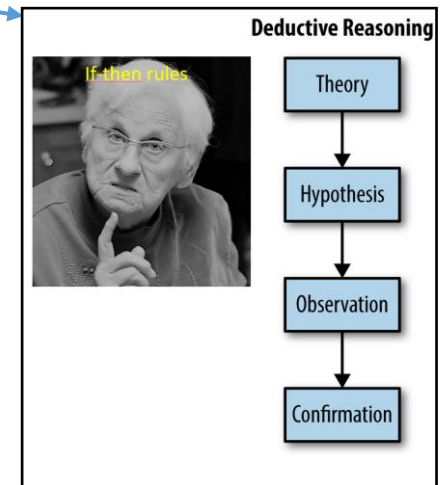
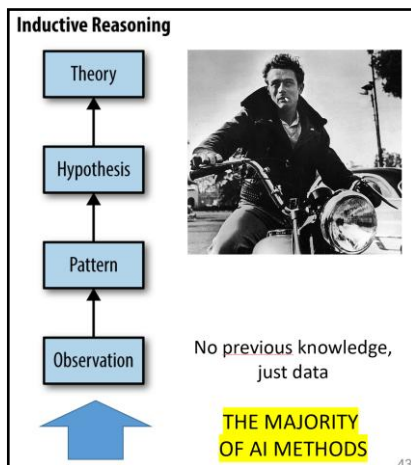
- known algorithms are too complex or inefficient

THEN

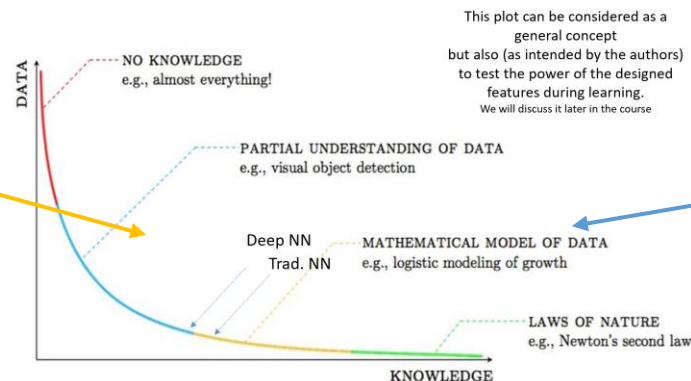
- AI can be considered as a possible solution

ELSE

- Standard coding should be considered as the first option



Data-Knowledge Spectrum



Classical design choices

- **Classical Coding** (e.g., coding the rules)
- Apply a trained **Model with ML**
- **Hybrid** approach
 - “Dividi et Impera”
 - some modules will be implemented by classical coding
 - some other modules with ML trained models

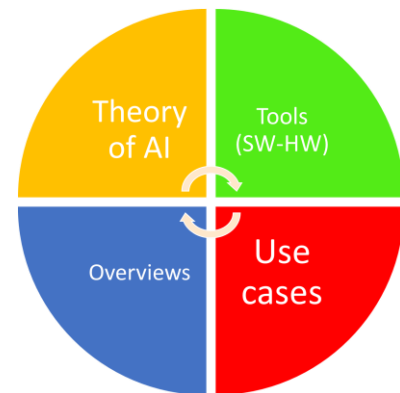




THEORY

White or Black box?

What level of knowledge do I need of my model for safe deployment?



Black/Gray/White box approach in AI

- This concept is already present in Software Engineering, but it fits perfectly also AI models



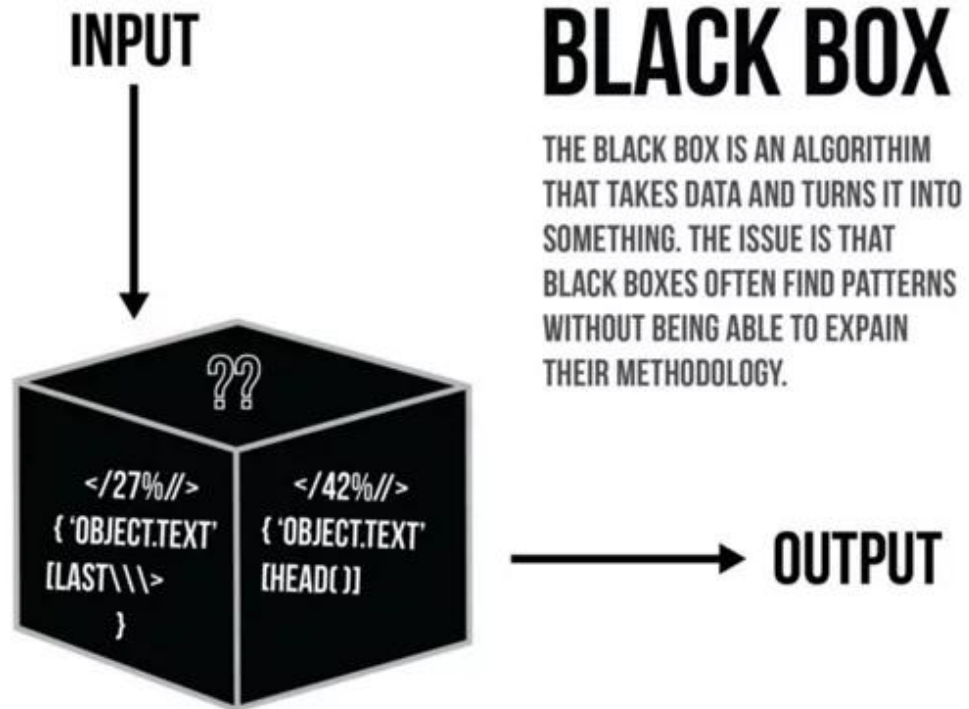
Black box approach

SAFE RULE

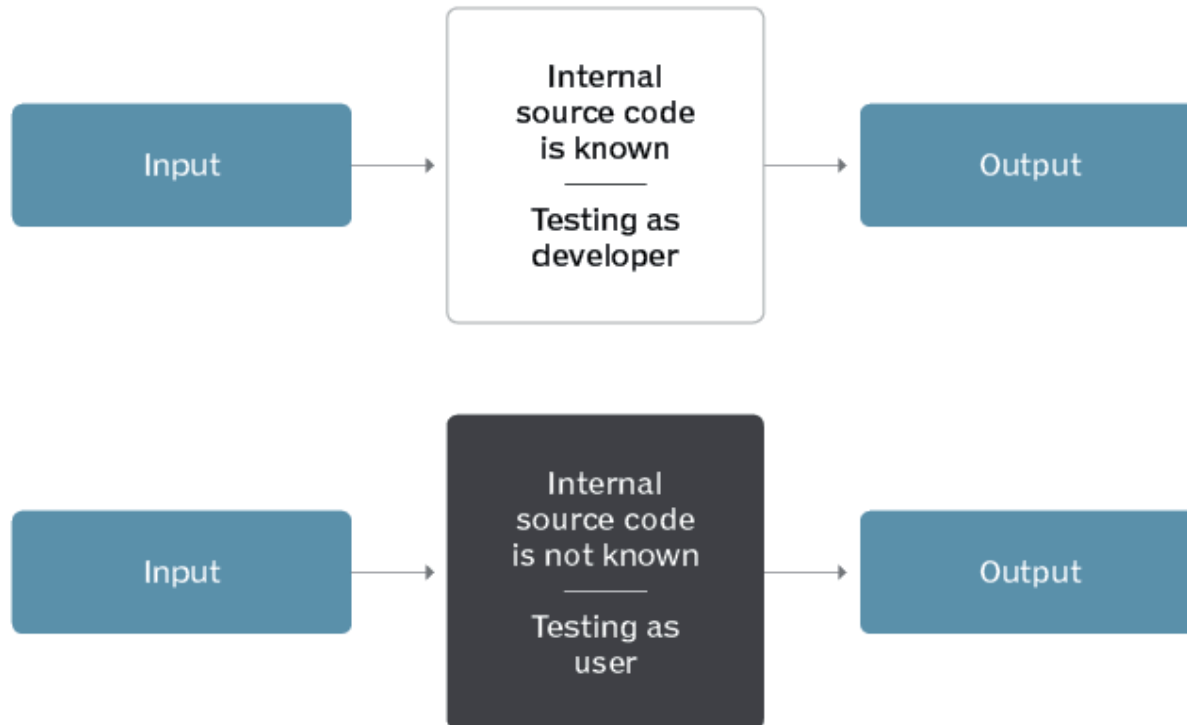
If you download and use a model that you can't understand how it is working

→ For YOU it's a black box

→ You have to manage it as a black box



White box testing vs. black box testing



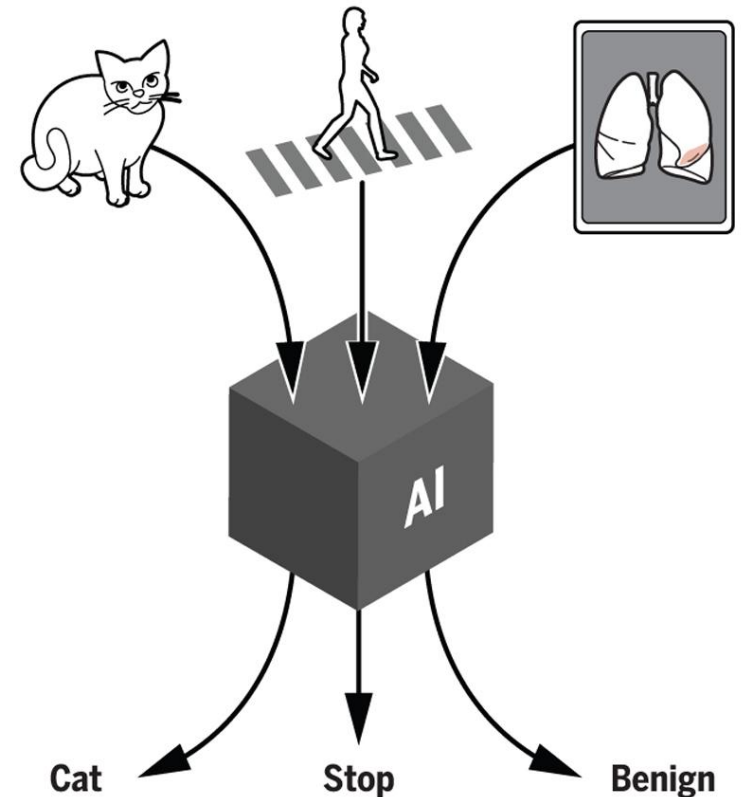
Maybe you understand the model,
but the number of parameters is so high
that is impossible to really understand the behavior

Can I use a black box?

Yes, IF...

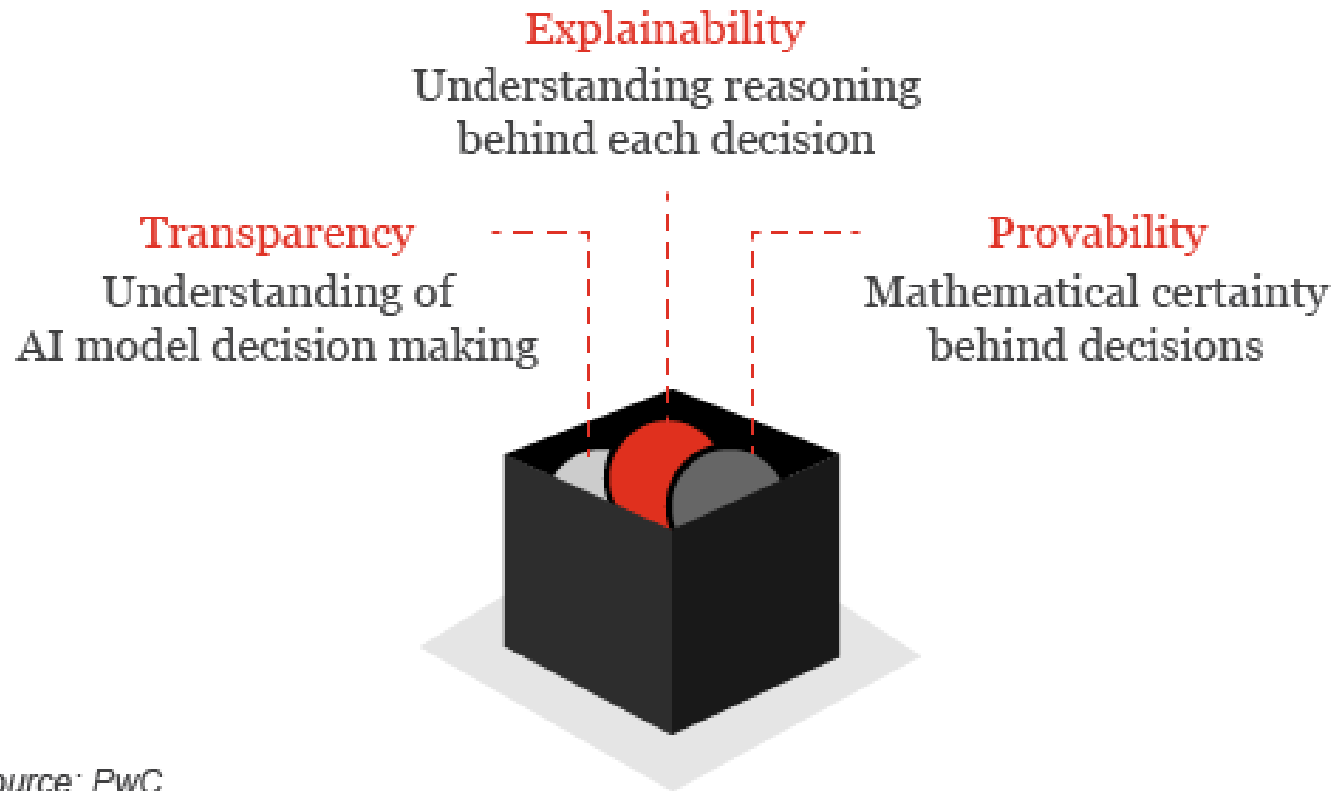
By definition, humans cannot assess how a black box algorithm arrives at a particular answer. However, black box methods can still provide value when they produce the best results, when the cost of a wrong answer is low, or when they inspire new ideas.

- yes, if risk is low
- yes, if output can be validated externally
- yes, if fallback procedures exist
- no, in high-risk critical applications without controls



You might also try to look inside the black box

What it means to look inside the black box



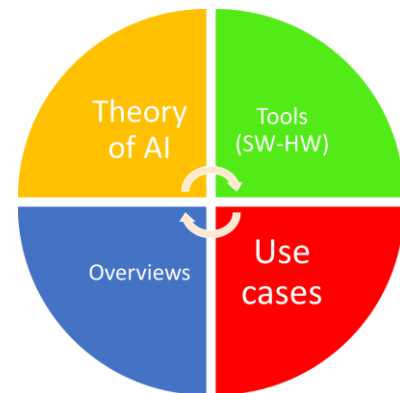
Source: PwC



THEORY

Size matters

What is the correct size of a ML model?



Sizes of “old” models: GPT

- Generative Pretrained Transformer 3 (GPT-3), developed by OpenAI in 2020, is a natural language processing (NLP) model
- GPT-3 produces articles, generated summaries of entire books, autocompleted codes, etc.
 - GPT-2, has only approximately 1.5 billion parameters and is trained on nearly 40 GB (giga) of text data
 - GPT-3 has roughly 175 billion parameters and is trained on nearly 45 TB (Tera) of text data

Sizes of recent models

(2023–2025)

- GPT-4 (OpenAI, 2023):
Estimated to have **1 trillion** parameters or more (exact figures are not publicly disclosed, but it's significantly larger than GPT-3).
Multimodal inputs (text, images) and advanced reasoning.
- Grok 3 (xAI, 2025):
Reported to have around 500 billion to **1 trillion** parameters, depending on configuration, with enhanced reasoning and real-world interaction capabilities.
- Gemini Ultra (Google, 2024):
Estimated at **1.5 trillion** parameters, designed for multimodal tasks and advanced problem-solving.

Trend

Model sizes have grown exponentially, but recent research suggests diminishing returns on scaling alone, with a focus on efficiency, data quality, and architecture (e.g., mixture-of-experts, sparse models).

Some other recent models

Anthropic	Anthropic	Anthropic	OpenAI Mar 2026	OpenAI
Claude Opus 4.6	Claude Sonnet 4.6	Claude Haiku 4.5	GPT-5.4	GPT-5.4 mini
claude-opus-4-6	claude-sonnet-4-6	claude-haiku-4-5	gpt-5.4	gpt-5.4-mini
CONTEXT 1M	CONTEXT 1M	CONTEXT 200k	CONTEXT 1.05M	CONTEXT 200k
128k out	64k out	64k out	272k std / 1M API	standard
INPUTS Text Image Doc/PDF	INPUTS Text Image Doc/PDF	INPUTS Text Image Doc/PDF	INPUTS Text Image Computer use	INPUTS Text Image
COMPLEXITY ★★★★	COMPLEXITY ★★★★★	COMPLEXITY ★★★★★	COMPLEXITY ★★★★★	COMPLEXITY ★★★★★
SWE-bench 80.8%	SWE-bench ~72%	Speed Fastest	OSWorld 75.0%	Speed Fast
GPQA Diamond 79.6%	Dev preference 70%	Cost/MTok in \$1	MMMU Pro 81.2%	Cost/MTok in \$0.75
GDPval 83.0%			GDPval 83.0%	
Deep reasoning, agentic workflows, long-context analysis, multi-agent coordination \$5 / \$25 per MTok	Balanced speed + intelligence; production coding; daily API default \$3 / \$15 per MTok	Fast, low-cost, high-volume tasks; sub-agents; classification; edge API \$1 / \$5 per MTok	Frontier reasoning + coding + native computer use; professional knowledge work; agentic desktop automation \$2.50 / \$15 per MTok	High-volume, cost-sensitive workloads; classification; RAG retrieval; sub-agents \$0.75 / \$4.50 per MTok

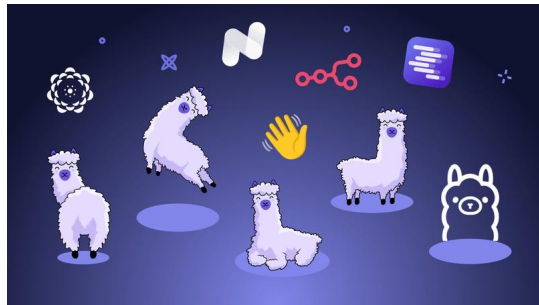
Inputs: Text Image Document / PDF Computer use (GUI) Complexity: ★★ basic ★★★ advanced ★★★★ frontier

Note: GPT-5.4 audio/video input not supported per official API docs (text + image only). Computer use is a native agentic tool, not a raw input modality. All models: text output only. Sources: Anthropic docs, OpenAI API docs — April 2026.

Far from the complexity of the human brain which has about **100 trillion parameters**

<https://www.computer.org/csdl/magazine/co/2022/05/09771130/1DeEYd2FXZm>

Some other local models

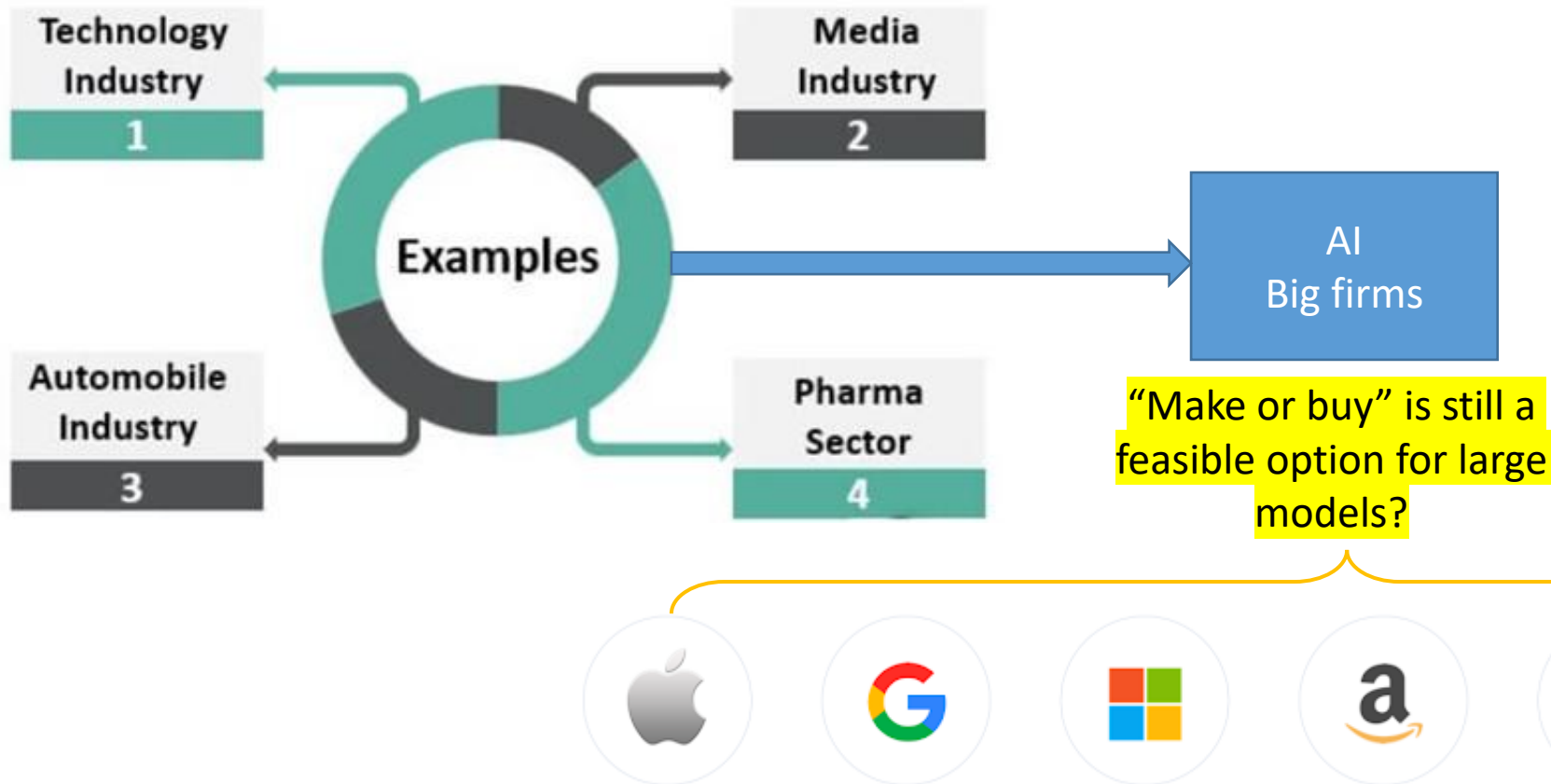


Open-source / local — cloud or self-hosted (2025-2026)						
Qwen3-VL 235B Alibaba — MoE	Open (Apache 2.0)	235B total 22B active	Text Image Video Document	128k tokens	★★★★★	Rivals Gemini 2.5 Pro on multimodal benchmarks; OCR, grounding
Qwen3.5-Omni Alibaba — March 2026	Open (Apache 2.0)	27B (dense)	Text Image Audio Video	128k tokens	★★★★★	Only open model with real-time streaming speech output; voice agents
Gemma 4 31B Google — 2026	Open (Google ToS)	31B dense	Text Image Video Audio (E2B/E4B)	256k tokens	★★★★★	MMMU Pro 76.9%; SigLIP encoder; edge-to-cloud; best math (AIME 89.2%)
Gemma 4 26B MoE Google — 2026	Open (Google ToS)	26B total 3.8B active	Text Image Video	256k tokens	★★★★☆	Most parameter-efficient; ranked 6th Arena AI; good for VRAM-limited setups
Llama 4 Scout Meta — April 2025	Open (Llama 4 Lic.) ⚠ no EU multimodal	109B total 17B active	Text Image	10M tokens	★★★★☆	Largest context of any open model; limited modality vs competitors
Llama 4 Maverick Meta — April 2025	Open (Llama 4 Lic.) ⚠ no EU multimodal	400B total 17B active	Text Image ↓	1M tokens	★★★★★	Highest MMLU (85.5%); datacenter GPU required; EU license

Far from the complexity of the human brain which has about 100 trillion parameters

<https://www.computer.org/csdl/magazine/co/2022/05/09771130/1DeEYd2FXZm>

AI oligopoly?



The larger the better?

In some case... NO!

Some very large model appear to be less narrow than previous models (e.g., 1T+ parameters) but.....

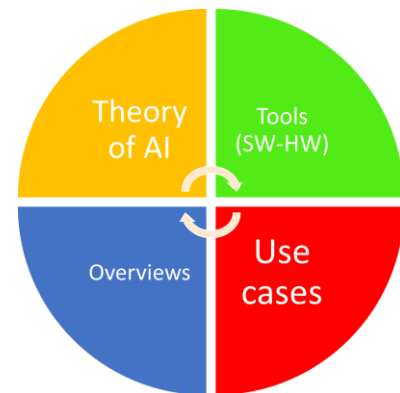
- A **stand-alone** device can't benefit of large models in the cloud
- **Size, power consumption, real-time capability** limit the applicability of a very large model
- Limited **explainability** of the large models exclude them to be applied in critical applications (see EU rules, Lesson 01)
- **Optimization, Quantization, Scalability, are key in many specific application**



THEORY

Machine learning basics

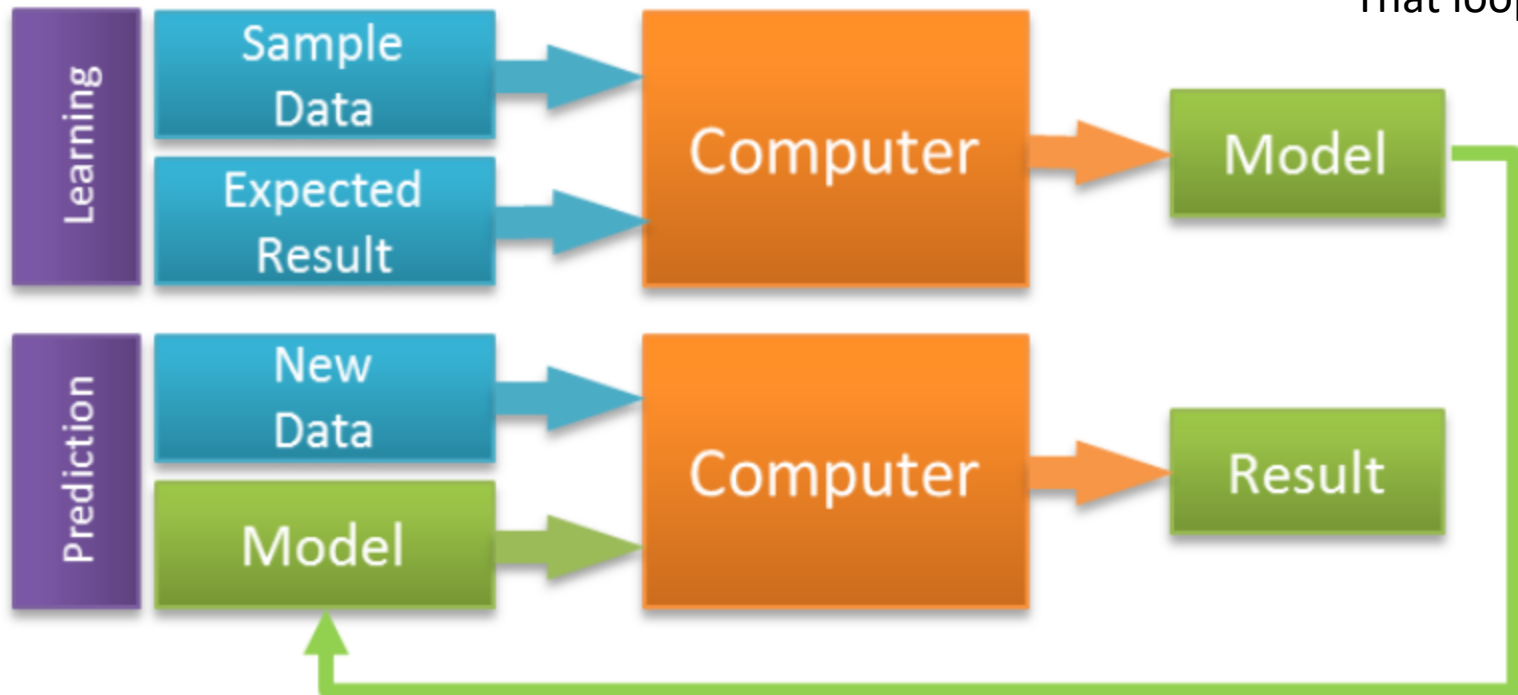
Model, Training, Evaluation



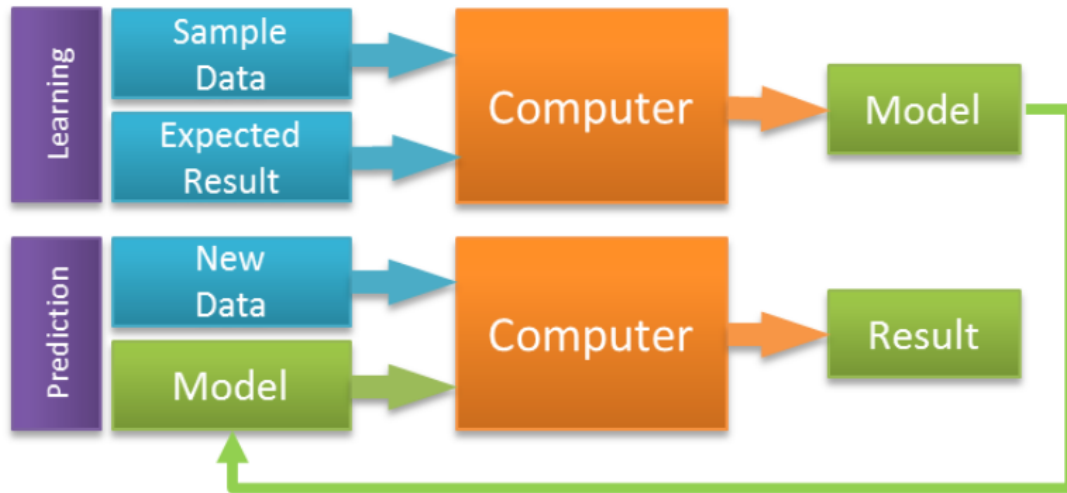
Traditional modeling:



Machine Learning:



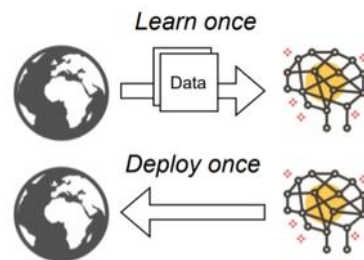
Machine Learning:



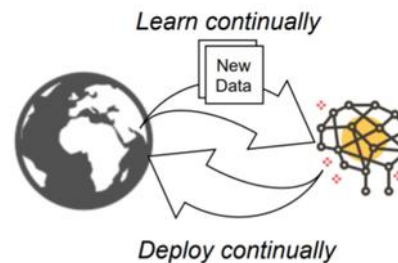
This loop can be done

- 1 time during development
- every week/day/hour in continual learning application

Static ML



Adaptive ML



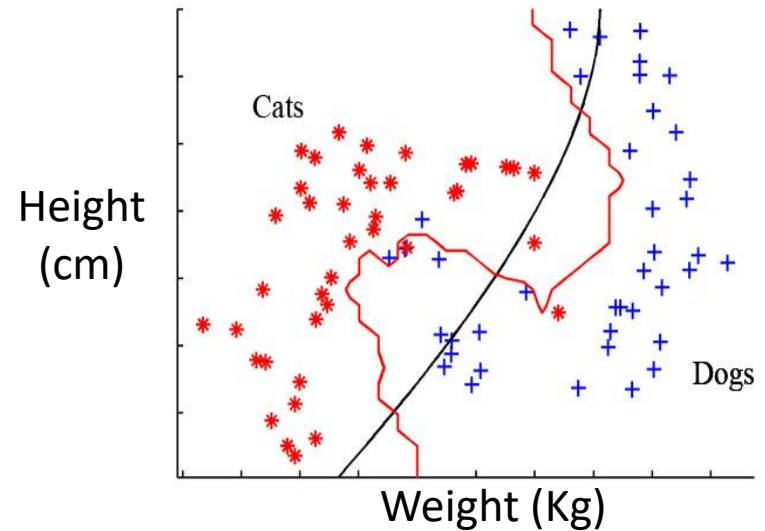
Example: Dog/Cat classifier

- **Classical programming**

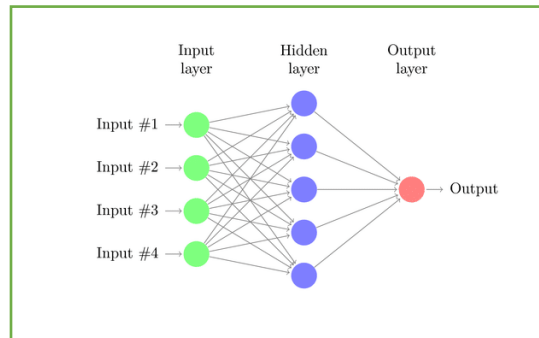
- IF $((W > A) \text{ AND } (H < B)) \rightarrow \text{dog}$
- IF $((A * W + B * H + C) > Z) \rightarrow \text{dog}$

- **ML**

- Table of classified data
- Train a model $Y = F(X)$
- Dog = IF $(F(X) > 0)$



$F() \equiv$



Model

ML in a metaphor

It's like gardening

- **Seeds** = Algorithms
- **Nutrients** = Data
- **Gardener** = You
- **Plants** = Programs



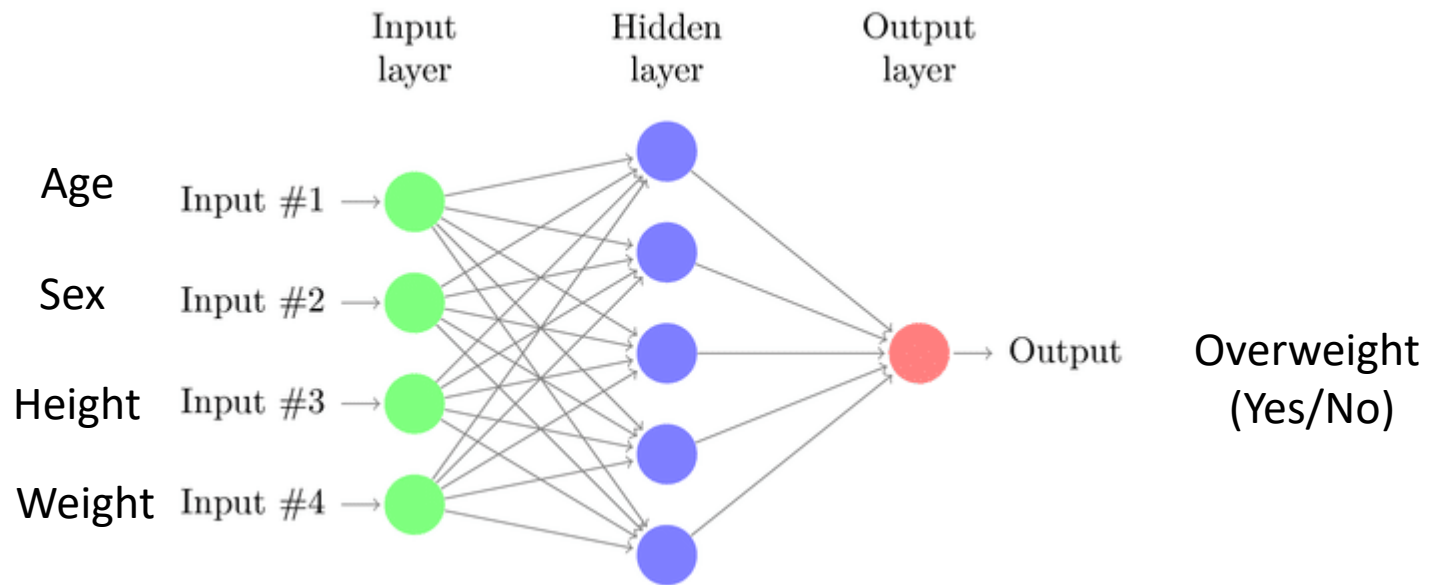


So... What Is Machine Learning?

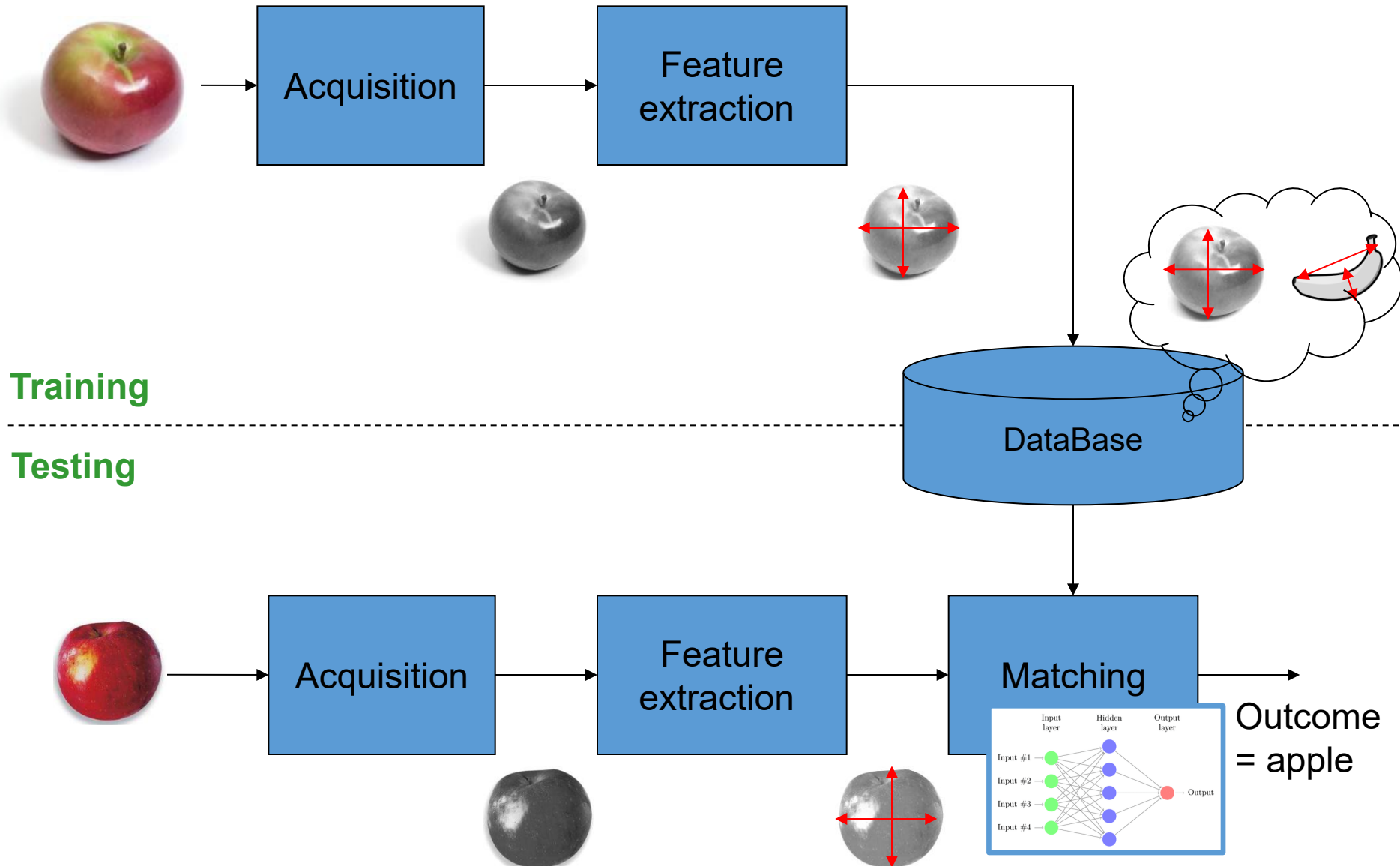
- Automating automation
- Getting computers to program themselves
- Writing software is the bottleneck
- Let the data do the work instead

Example: Neural Network

- Without lack of generality in the following we will refer to the feed-forward NNs

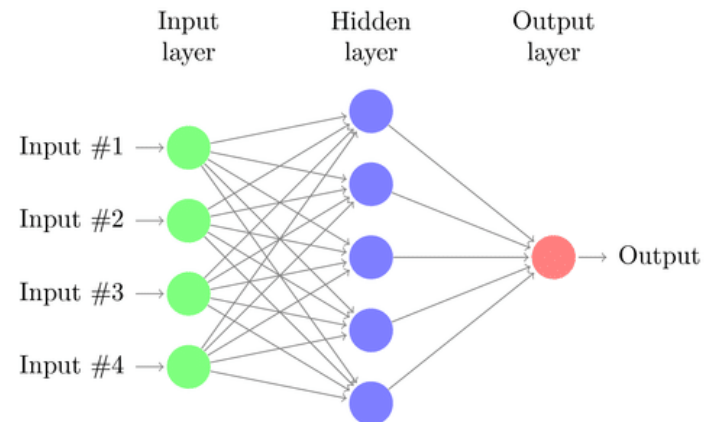


Pattern Recognition Systems

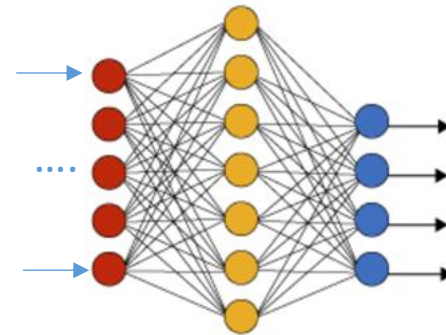


ML: the 3 main components

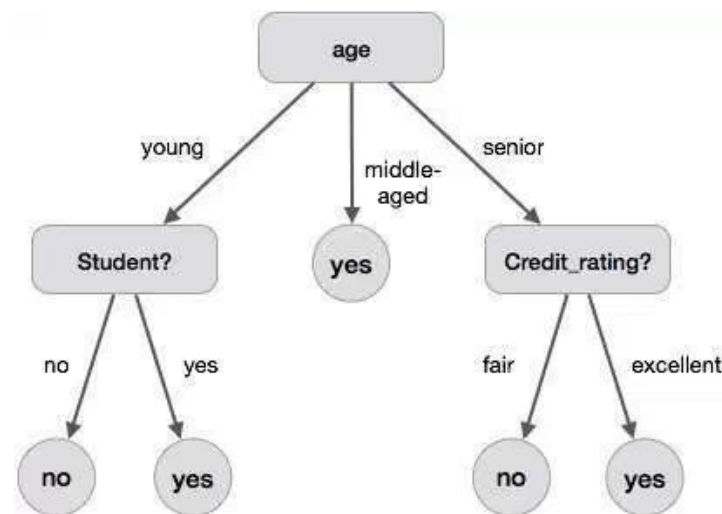
- Tens of thousands of machine learning algorithms. Hundreds new, every year
- Every machine learning algorithm has **3 components**:
 - Representation
 - Evaluation
 - Optimization



Representation



- Decision trees
- Sets of rules / Logic programs
- Instances
- Graphical models (Bayes/Markov nets)
- Neural networks
- Support vector machines
- Model ensembles
- Etc.



About Representation: we have different features

Which features is relevant for your application?

(This is not the complete list)



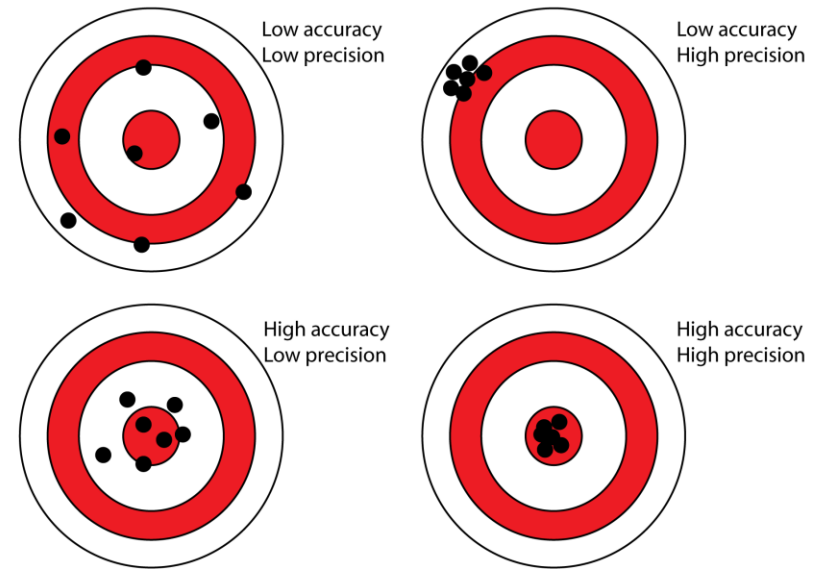
AI Method	Distributed	Real-Time	Embedded	Robust	Autodidactic	...
Artificial Neural Network	✓	✓	✓	✓	✗	
Bayesian Network (dynamic structure)	✓	✗	✓	✗	✗	
Bayesian Network (static structure)	✓	✓	✓	✗	✓	
Expert System	✓	✗	✗	✗	✓	
Reinforcement Learning	✓	✗	✓	✗	✗	
Random Forest	✓	✓	✓	✓	✗	
Support Vector Machine	✓	✓	✓	✓	✗	

Autodidactic:

should support being updated with
new information as it becomes available.

Evaluation

- Accuracy
- Precision and recall
- Squared error
- Likelihood
- Posterior probability
- Cost / Utility
- Margin
- Entropy
- K-L divergence
- Etc.

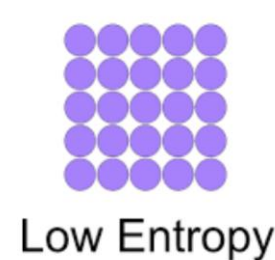
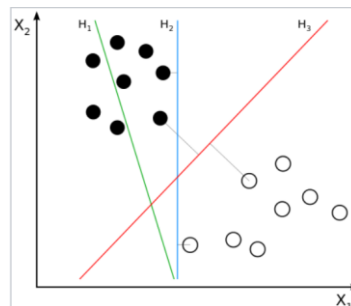


$$P(c|x) = \frac{P(x|c)P(c)}{P(x)}$$

Labels for the equation components:

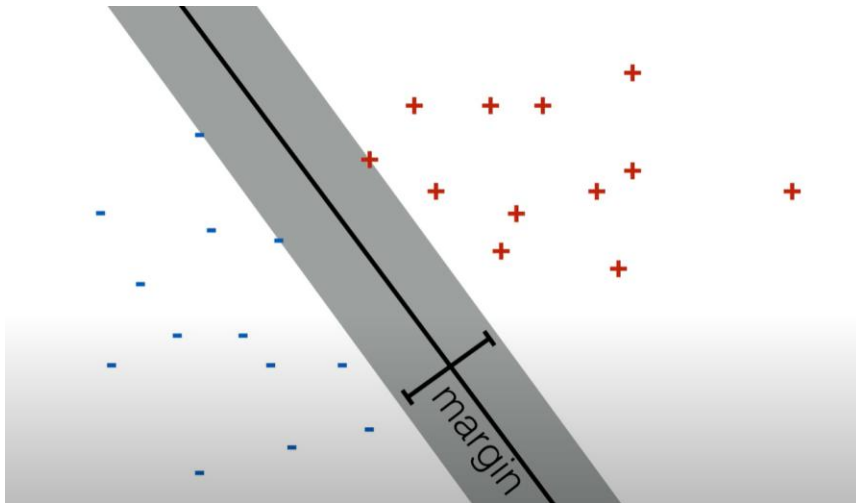
- $P(x|c)$: Likelihood
- $P(c)$: Class Prior Probability
- $P(c|x)$: Posterior Probability
- $P(x)$: Predictor Prior Probability

Having in input x , which is the probability that is a dog (Hypot. c)?



Evaluation: comparison between two figures

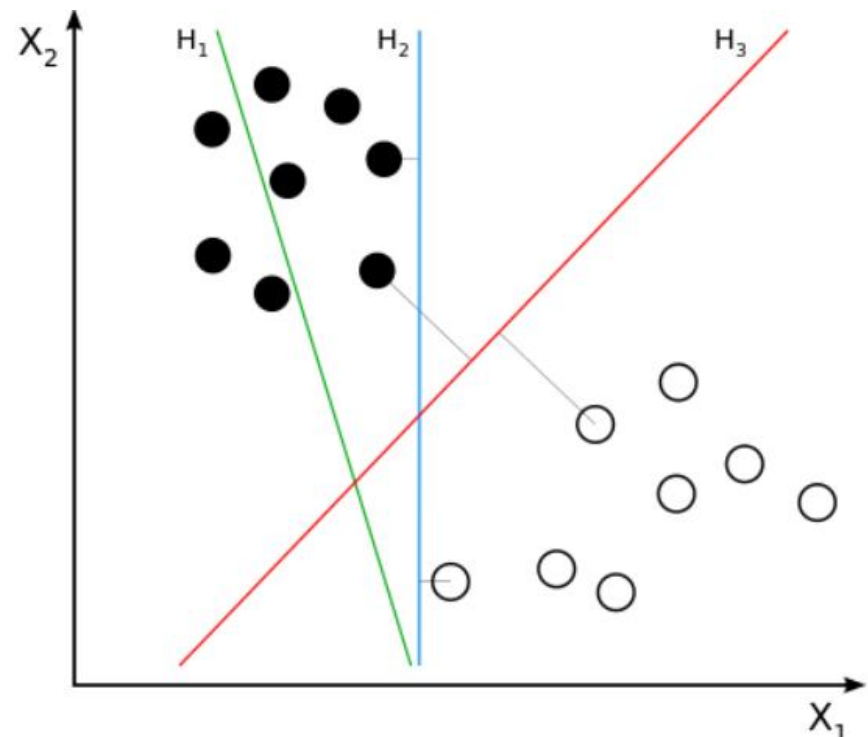
Accuracy VS Margin



The decision boundaries H_2 and H_3 have both 100% accuracy (error = 0), but the margin values are different

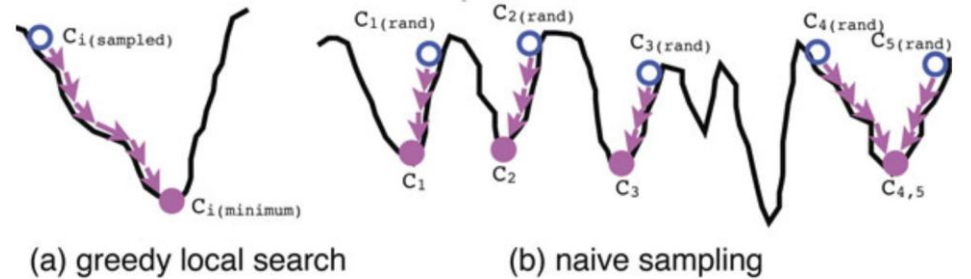
We want to augment the margin!

We prefer H_3

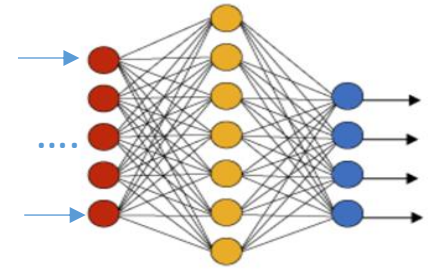
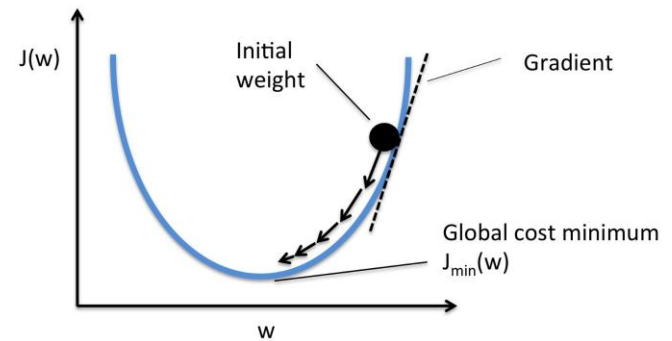


Optimization

- Combinatorial optimization
E.g.: Greedy search



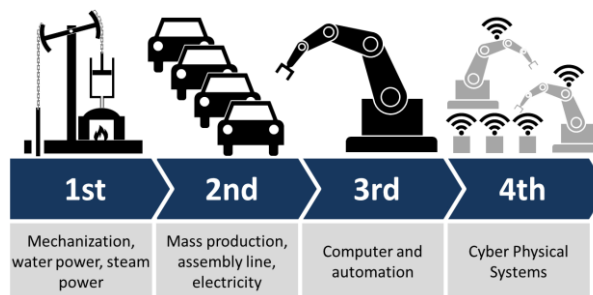
- Convex optimization
E.g.: Gradient descent
- Constrained optimization
E.g.: Linear programming





Overview

Digital ages and recent challenges



A digital word:

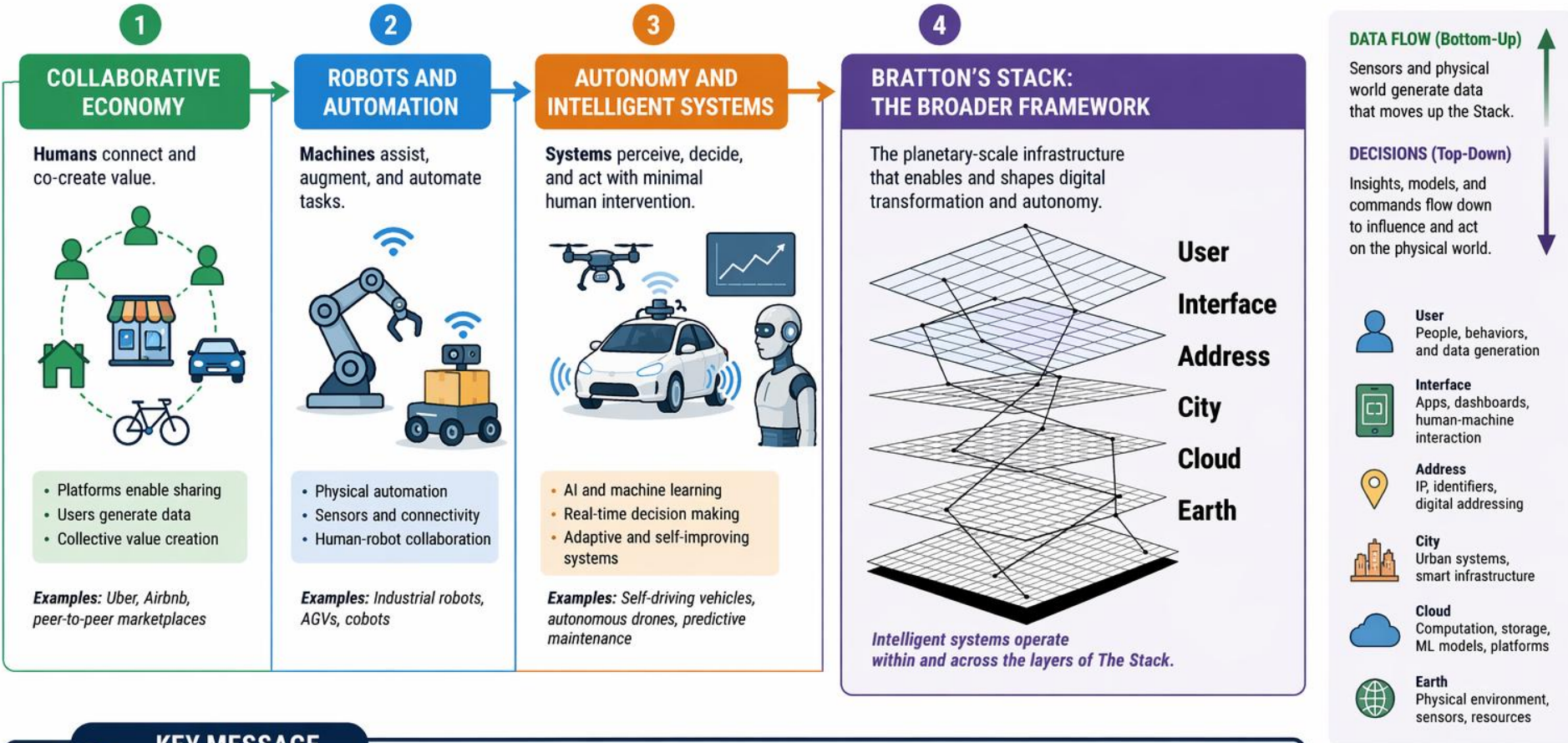
challenges for environment and industries

- Players are facing huge challenges but across the globe relevant initiatives are present for example:
 - Industry 4.0 and Industry 5.0
 - Internet of Things (IoT)
 - Internet of Everything (IoE) - Cisco def.
- Artificial intelligence
 - increasing relevance in every application!

In the next slides we review the status and the trends of the current technology with some concepts and example

Digital Transformation: From Collaboration to Autonomy

A conceptual extension moving from collaborative economy to robots, autonomy, and intelligent systems, framed within Bratton's Stack – the global infrastructure that makes it possible.



KEY MESSAGE



Digital transformation is not just about technology.
It is an evolution from human collaboration to intelligent autonomy, enabled and shaped by a global computational infrastructure: The Stack.

AI, robotics, and autonomy do not replace The Stack – they thrive within it.

Autonomous systems in agriculture



AgBot 5.115T2



AgBot 2.055W3



AgBot 2.055W4

Spraying
Hoeing
Mulching
Tillage
Seeding
Weeding
Mowing
Swathing
Teding

Autonomous systems for elder people



Working in different houses/environments

Tidying

Cleaning

Perceiving and grasping transparent objects

The **Tactile team** at TRI is working on highly compliant tactile sensors and the algorithms to put them to good use

<https://punyo.tech/>



EXAM SIMULATION

To TEST your preparation
(do not use the simulation to study)



Please remember



Multiple-choice test:

- 1 single correct answer $\rightarrow weight = 1$
 - N wrong answers $\rightarrow weight = 0$
 - ~~1 (rarely 2) very bad answer $\rightarrow weight = -1$ (No more)~~
1. No penalties for not answering
 $\rightarrow$ OK BUT IT'S BETTER TO TRY TO ANSWER
 2. **DO NOT LOOK FOR THE RIGHT ANSWER, EXCLUDE THE WRONG ONES**
 3. DOUBTS? $\rightarrow$ Choose the answer that "looks less wrong" to you

.Q1



An AI model is processing an input RGB image to evaluate the age expressed in years of the face present in the image. What kind of model is it?

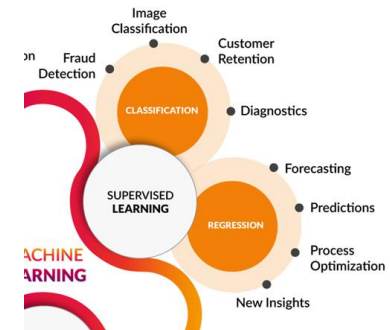
1. Classifier model
2. Regressor model
3. Clustering model
4. Reinforced Learning model
5. None of the above

.Q1



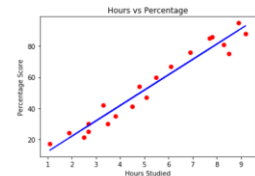
An AI model is processing an input RGB image to evaluate the age expressed in years of the face present in the image. What kind of model is it?

1. Classifier model
2. Regressor model
3. Clustering model
4. Reinforced Learning model
5. None of the above



Regression

- It predicts continuous valued output.
- Compute the probabilistic relationship between variables for the purposes of forecasting or prediction.



..Q2



Artificial Intelligence can be applied to the following sectors

1. Robotics
2. Information extraction
3. All the above

..Q2



Artificial Intelligence can be applied to the following sectors:

1. Robotics
2. Information extraction
3. All the above ← correct

Comment: the range of applications is really exploding including 1) and 2)

...Q3



Artificial neural networks are capable to learn human biases.

1. False: the achievable complexity of the artificial neural networks is so far from the complexity of the human brain to make impossible to mimic this characteristic
2. False: human biases are not reproducible nor measurable
3. True.

...Q3



Artificial neural networks are capable to learn human biases.

1. False: the achievable complexity of the artificial neural networks is so far from the complexity of the human brain to make impossible to mimic this characteristic
2. False: human biases are not reproducible nor measurable
3. True ← correct

Comment: it's true because the humans prepare the datasets, and the statistical distribution of the samples among different labels and classes can strongly influence the learning of the artificial intelligent models.

....Q4



Recent artificial intelligence models can solve analogy puzzles like

“Paris is to France as Tokyo is to ?”
producing the correct answer “Japan.”

1. True
2. False

....Q4



Recent artificial intelligence models can solve analogy puzzles like
“Paris is to France as Tokyo is to ?”
producing the correct answer “Japan.”

1. True ← correct
2. False

Comment: yes. So-called “word embeddings” that are often used as input for a neural network are learned from large text collections and are able to solve analogy puzzles.

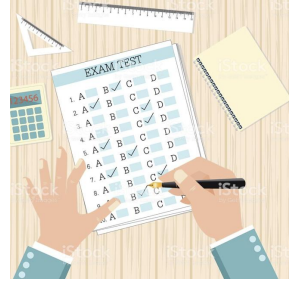
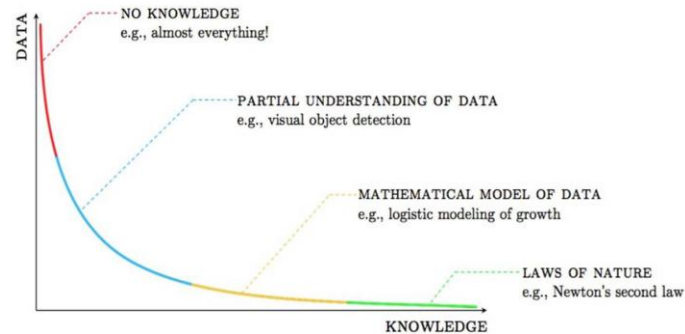
.....Q5



Considering the “Data knowledge spectrum plot” discussed in class, the **minimum** amount of data required is in the following case.

1. No knowledge about the model generating the data is available
2. A statistical model of the process is available
3. A mathematical model of the process is available

.....Q5



Considering the “Data knowledge spectrum plot” discussed in class, the **minimum** amount of data required is in the following case.

1. No knowledge about the model generating the data is available
2. A statistical even is limited model of the process is available
3. A mathematical model of the process is available ← correct

Comment: since just some parameters of the mathematical model must be tuned/fitted, the number of needed data is less than the amount needed to train the complete model with no a-priori information.

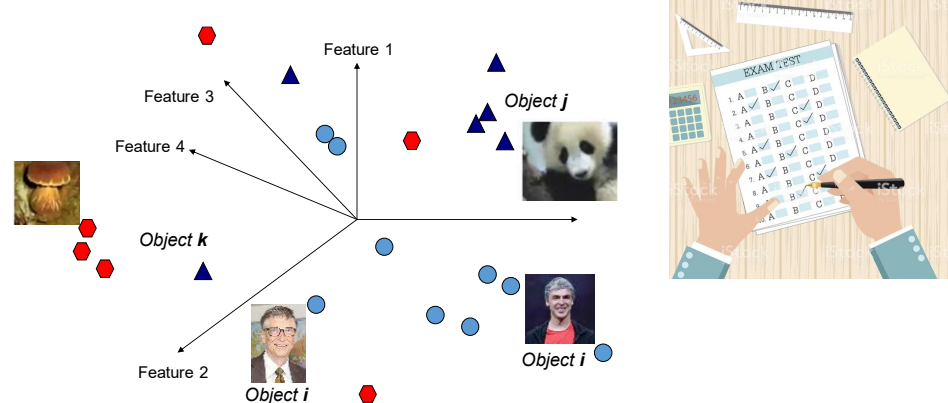
.....Q6



It is possible to think to the single datum in input to the neural network as a point in the “input space” of the model, even if the input is a single value, a N dimensional vector, or an image

1. True
2. False

.....Q6



It is possible to think to the single datum in input to the neural network as a point in the “input space” of the model, even if the input is a single value, a N dimensional vector, or an image

1. True ← correct

2. False

Comment: yes, that is exactly the correct representation that will allow to understating under a common framework the artificial intelligence models

.....Q7



It is correct to say the one of the key feature of an intelligent artificial system is the capability to learn (even if only a limited sense) and/or get better in time

1. True
2. False

.....Q7



It is correct to say the one of the key feature of an intelligent artificial system is the capability to learn (even if only a limited sense) and/or get better in time

1. True ← correct

2. False

Comment: One can debate endlessly about whether a certain system is intelligent or not, but the capability to learn is one of the most important

.....Q8



According to the Andries Engelbrecht definition of Computational intelligence what of the following is not included?

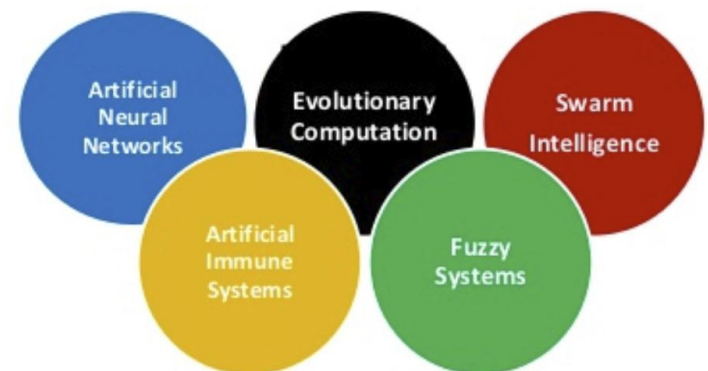
1. Artificial Neural Networks
2. Evolutionary Computing
3. Swarm Intelligence
4. Artificial immune system
5. Fuzzy Systems
6. All the above are included

.....Q8



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1. Artificial Neural Networks
2. Evolutionary Computing
3. Swarm Intelligence
4. Artificial immune system
5. Fuzzy Systems
6. All the above are included ← correct



.....Q9



According to the class discussion of the Gestalt capability, what of the following sentences is more correct?

1. The Gestalt capability is a typical feature present by-design in the model of classical neural networks
2. The Gestalt capability is a typical feature present by-design in the model of deep learning neural networks
3. The Gestalt capability is a typical human feature not well (yet) mimicked in current artificial networks

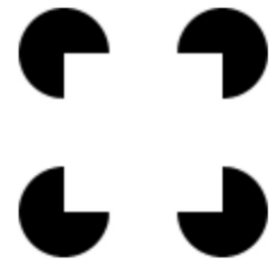
.....Q9



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← correct

Comment: just let's think to this example...



.....Q10



The following activity:

- a) Data Selection;
- b) Data Filtering;
- c) Data Enhancing...

1. Are part of the job of the artificial intelligent specialist in normal activities
2. Contribute to keep lower the complexity of the learning task
3. All the above (1. and 2.)
4. Are part of the classical machine learning approaches and they are (correctly) no longer used in deep learning applications

.....Q10



The following activity:

- a) Data Selection;
- b) Data Filtering;
- c) Data Enhancing...

1. Are part of the job of the artificial intelligent specialist in normal activities
2. Contribute to keep lower the complexity of the learning task
3. **All the above** (1. and 2.) **← correct**
4. Are part of the classical machine learning approaches and they are (correctly) no longer used in deep learning applications

Comment: a), b) and c) are extremely important in the final behavior of the trained model and the complexity of the training task.

.....Q11



The Mean Squared Error is typically present in what step of the design?

1. Representation
2. Evaluation

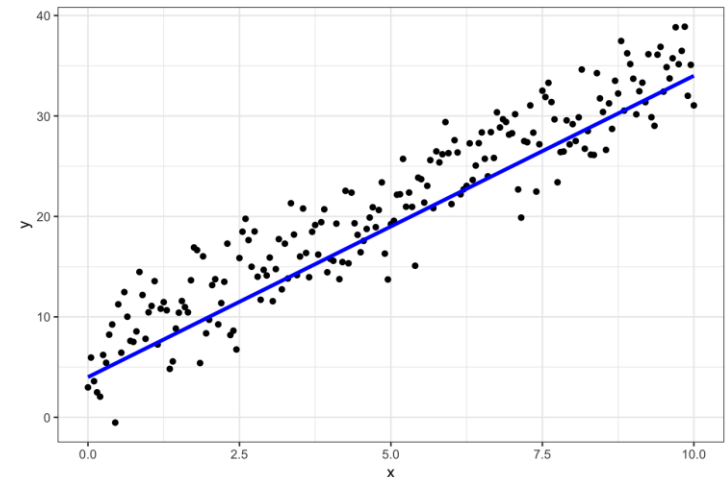
.....Q11



The Mean Squared Error is typically present in what step of the design?

1. Representation
2. **Evaluation**

$$MSE = \frac{1}{n} \sum \underbrace{\left(y - \hat{y} \right)^2}_{\substack{\text{The square of the difference} \\ \text{between actual and} \\ \text{predicted}}}$$



.Q12



Nowadays, the usage of *classical feature extraction* and *data analysis methods* is outdated since the capability of the recent deep learning models and methods made them obsolete and not more present in the common practice.

1. True
2. False

.Q12



Nowadays, the usage of *classical feature extraction* and *data analysis methods* is outdated since the capability of the recent deep learning models and methods made them obsolete and not more present in the common practice.

1. True
2. False ← correct

Comment: most part of the job is about prepare, study and validate the data to create efficient datasets, also with classical tools!

What about topics not present in the simulation?

- They are equally important
 - The absence in the simulation does not mean they are not relevant
- You can find them in the exam
- Do not study the simulation, but use it to understand if you are profitably attending the course and, in case, to change the study method
- **THE CORRECT METHOD:**
Try to create your own simulation by browsing all slides of the course and ask yourself:
what can be asked in this slide?



Main points



- Black box approaches
- Theory of intelligent systems:
Representation, *Evaluation* and *Optimization*
- The design is like gardening:
Seeds = Algorithms, *Nutrients* = Data,
Gardener = You, *Plants* = Programs
- Digital Ages
 - Collaborative age
 - Autonomous age
- Exam simulation

Annex

(Not part of the exam,
but **useful for generating new ideas**)

Digital ages and recent challenges

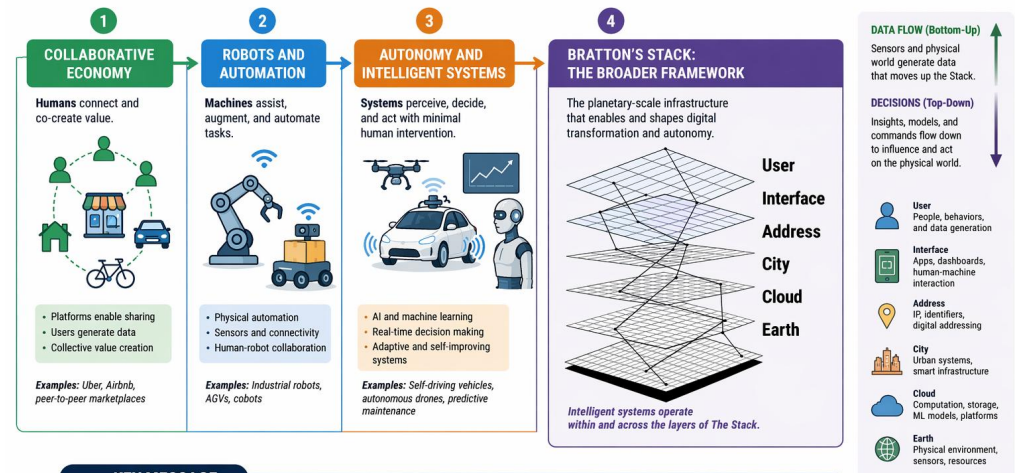


Annex description

This annex is a **conceptual extension on digital transformation**, moving from **collaborative economy** to **robots, autonomy, and intelligent systems**, ending with **Bratton's Stack** as a broader interpretive framework.

Digital Transformation: From Collaboration to Autonomy

A conceptual extension moving from collaborative economy to robots, autonomy, and intelligent systems, framed within Bratton's Stack – the global infrastructure that makes it possible.



KEY MESSAGE



Digital transformation is not just about technology. It is an evolution from human collaboration to intelligent autonomy, enabled and shaped by a global computational infrastructure: The Stack.

AI, robotics, and autonomy do not replace The Stack – they thrive within it.



Overview

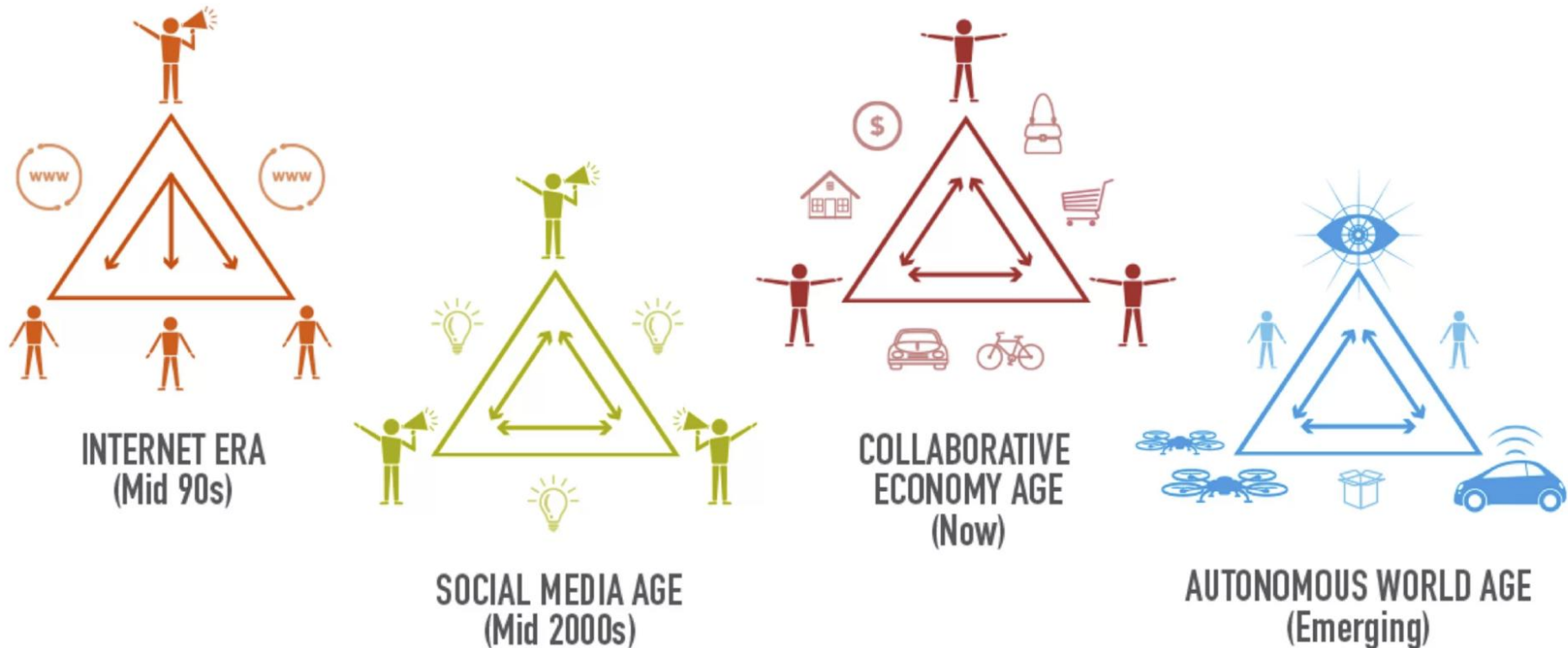
Digital ages and recent challenges



The 4 phases of the digital ages

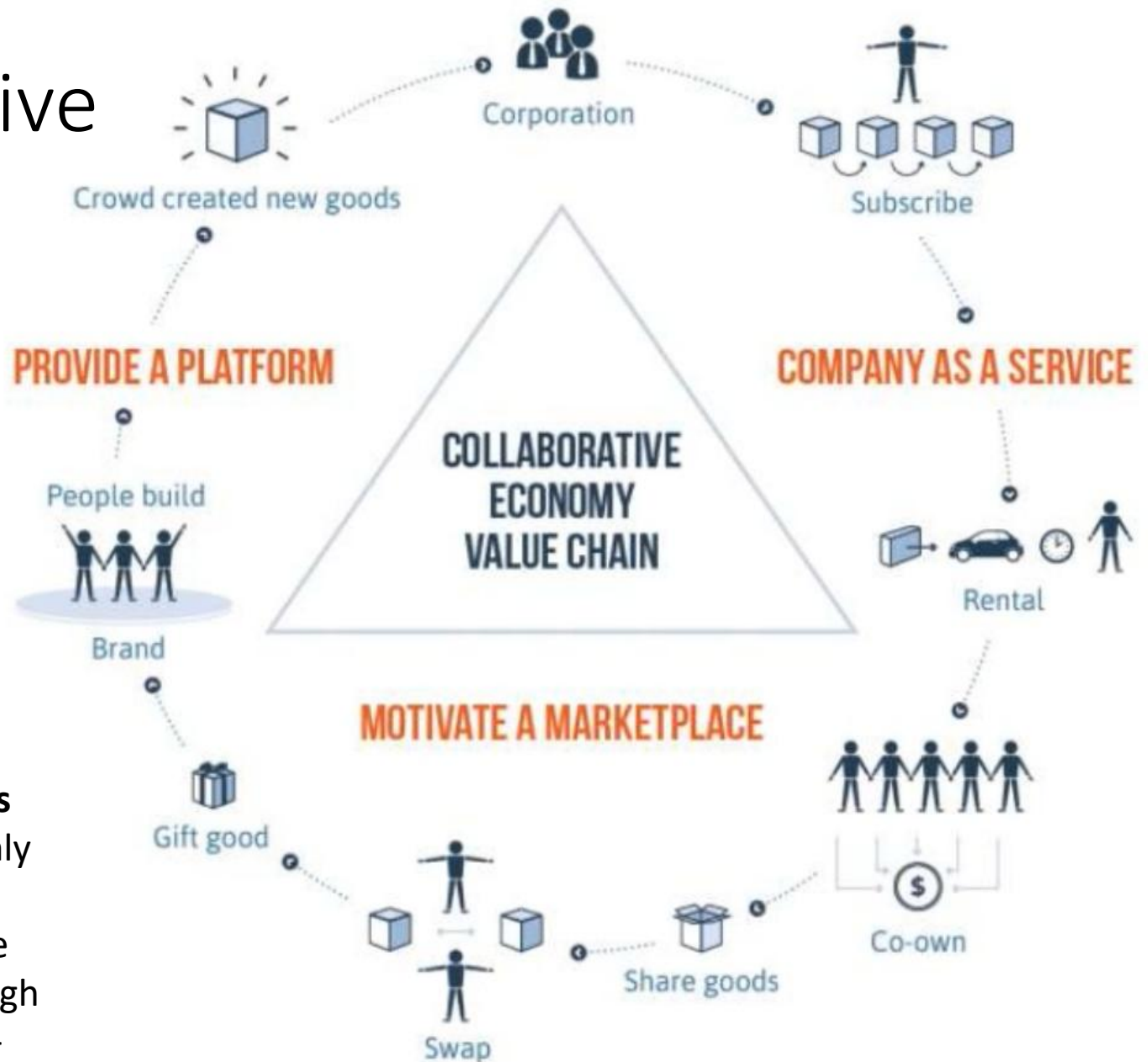
We are currently between the collaborative economy and the autonomous world

Today's systems are still largely human-centered, but increasingly supported by data-driven and AI-based decision processes. Full autonomy is emerging, but not yet fully deployed in real-world applications.



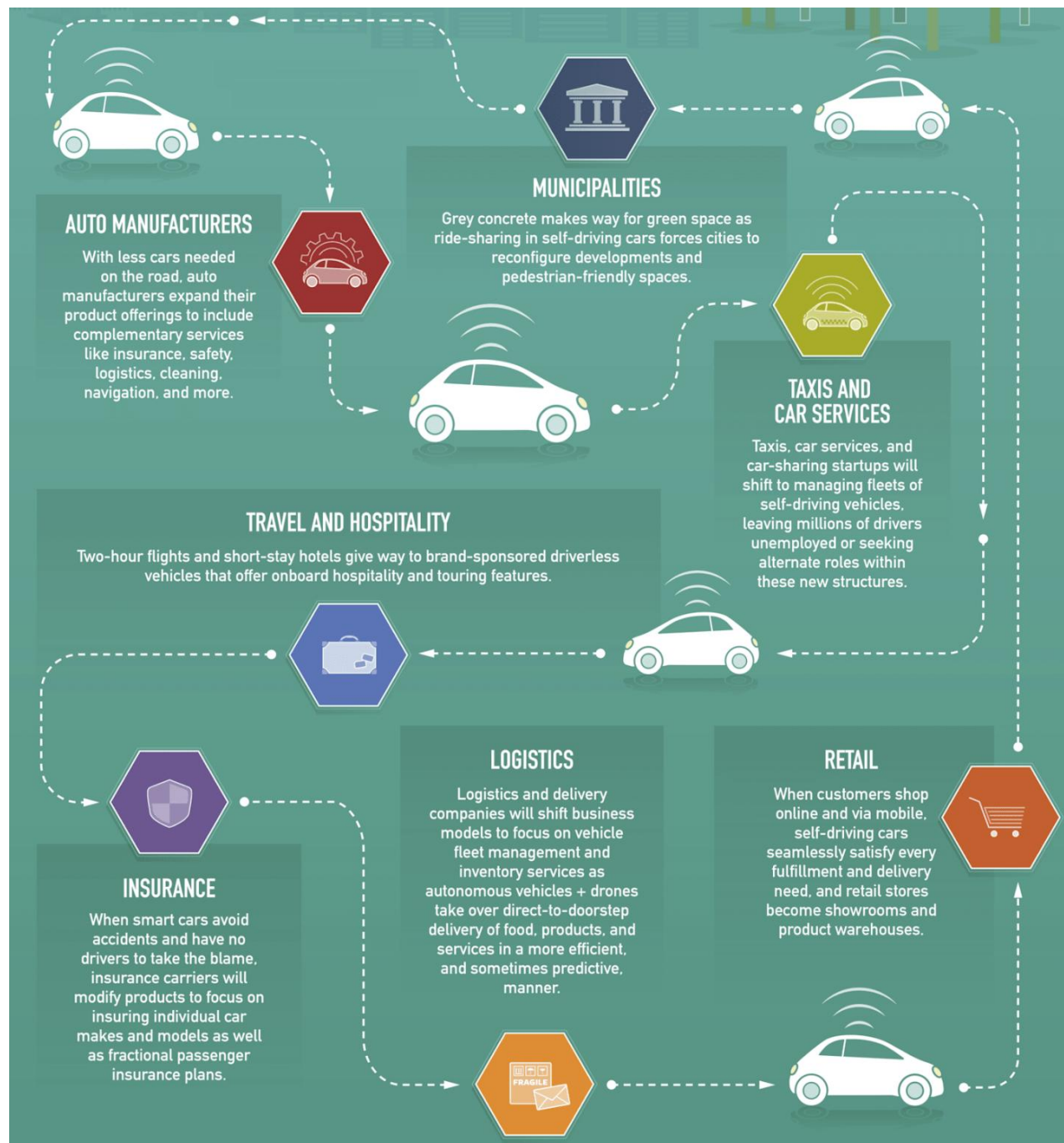
Autonomous world:
intelligent technology systems (operating without human participation)
enable new business models in a more efficient society.

Collaborative economy



The collaborative economy is based on platforms connecting users, data, and services. Companies no longer only produce goods, but provide platforms where users create value through sharing, renting, and co-owning resources.

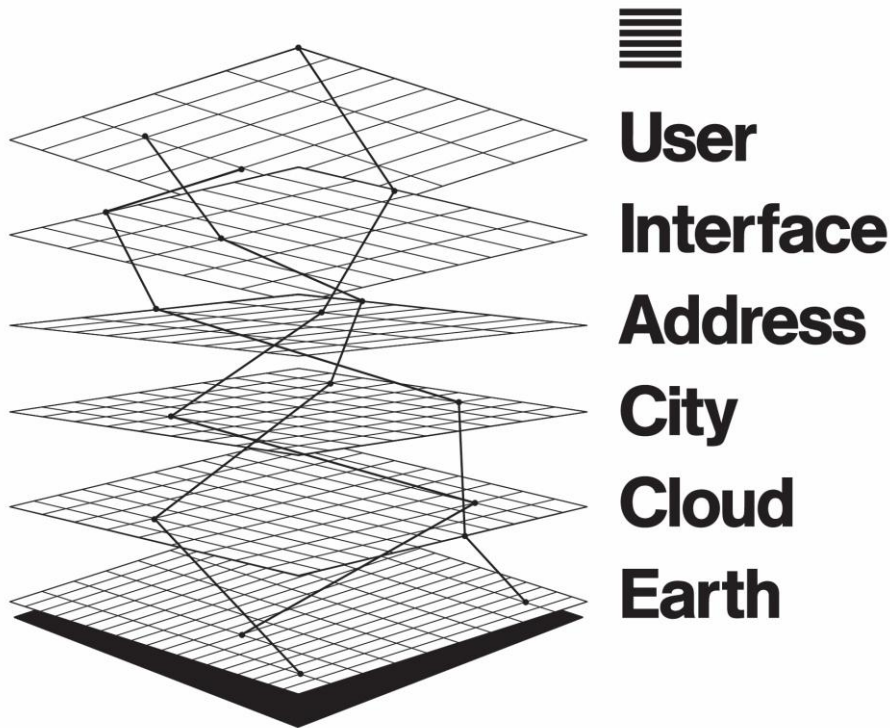
Robots + Autonom. + Int. Sys



The Bratton's stack



Benjamin H. Bratton
Sociologist



These different genres of computation — smart grids, cloud platforms, mobile apps, smart cities, the Internet of Things, automation— can be seen not as so many species evolving on their own, but as forming a coherent whole: an **accidental megastructure** called “The Stack” that is both a computational apparatus and a **new governing architecture**. We are inside “The Stack” and it is inside of us.

The world is now organized by a global computational system made of interconnected layers (The Stack), not just by countries or companies

What the diagram shows

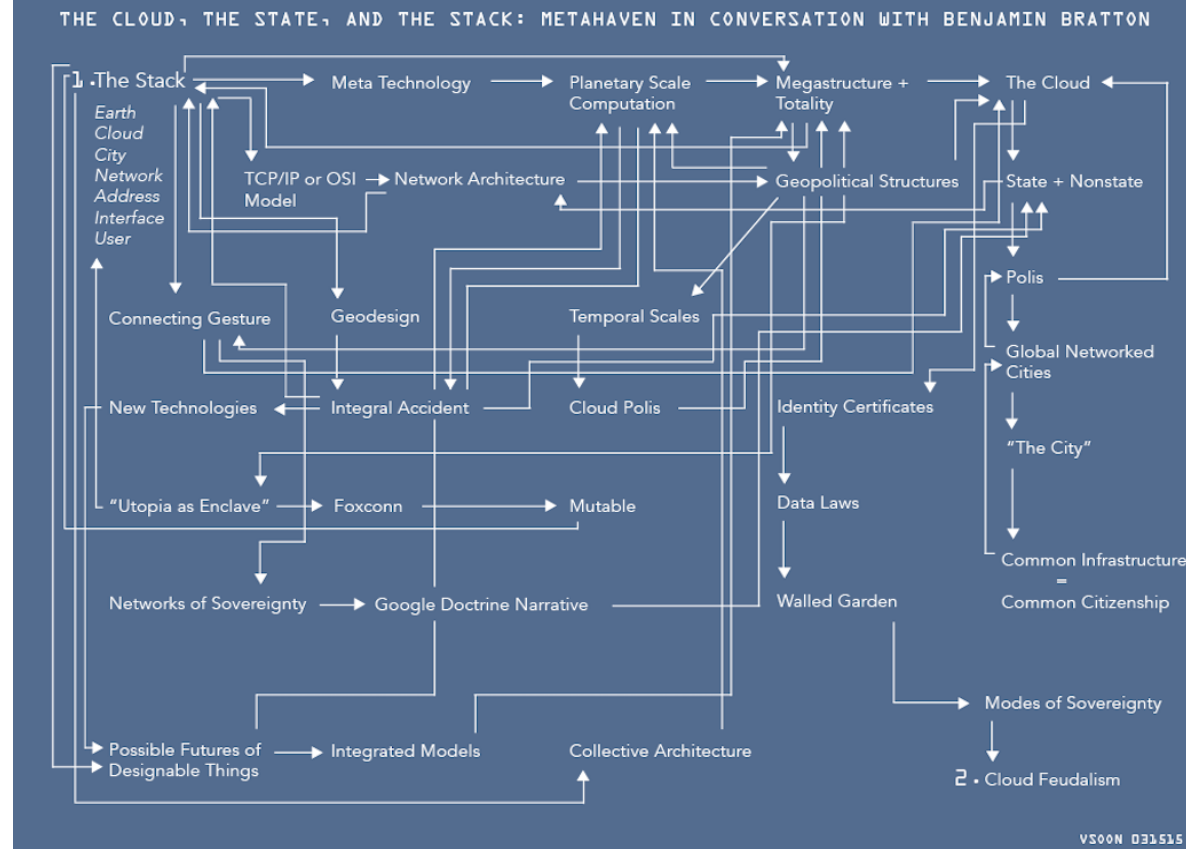
All the arrows mean:

Everything is interconnected

- technology ↔ politics
- data ↔ identity
- infrastructure ↔ economy

Examples in the diagram:

- Cloud → State + Nonstate
- Data laws → identity → geopolitics
- Platforms → marketplaces → cities



-> So:

**technology is shaping society,
not just supporting it**